

отчёт по лабораторной работе №3

Дискреционное разграничение прав в Linux. Два пользователя

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1 Цель работы

Получение практических навыков работы в консоли с атрибутами файлов для групп пользователей¹.

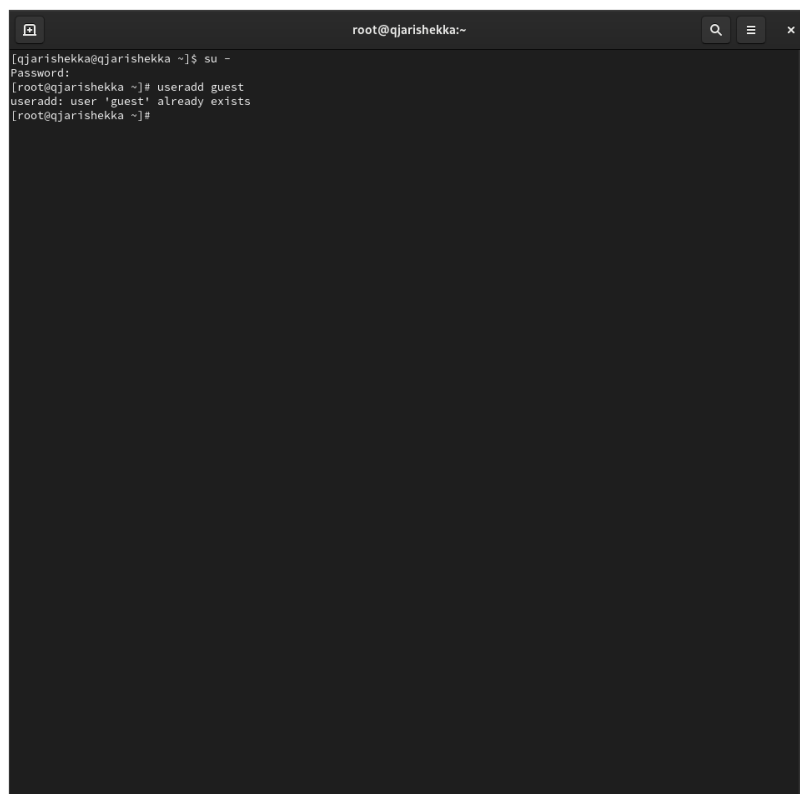
2 Задание

- Создание нового пользователя и изменение прав доступа.
- Заполнение таблицы

3 Выполнение лабораторной работы

Сначала я запустил терминал и выполнил команду для создания нового пользователя guest (рис. 3.1).

```
useradd guest
```

A screenshot of a terminal window with a dark background. The window title is 'root@qjarishekkka:~'. The terminal shows the following sequence of commands and output: a user prompt '[qjarishekkka@qjarishekkka ~]\$' followed by 'su -', a 'Password:' prompt, a root prompt '[root@qjarishekkka ~]#', the command 'useradd guest', the output 'useradd: user 'guest' already exists', and finally another root prompt '[root@qjarishekkka ~]#'.

```
root@qjarishekkka:~  
[qjarishekkka@qjarishekkka ~]$ su -  
Password:  
[root@qjarishekkka ~]# useradd guest  
useradd: user 'guest' already exists  
[root@qjarishekkka ~]#
```

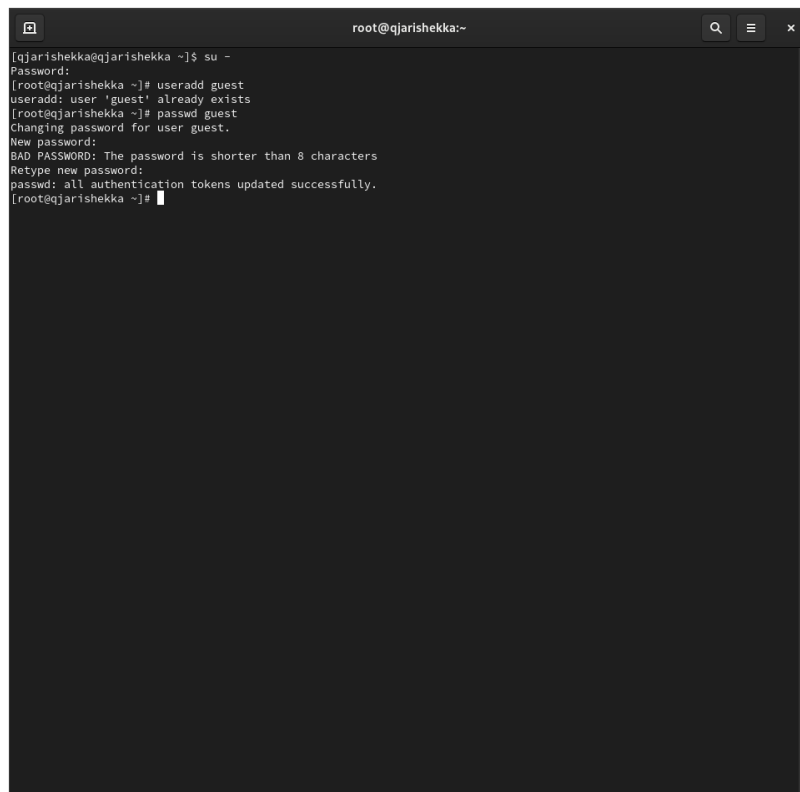
Рис. 3.1: создание нового пользователя

Потом я изменил пароль пользователя, я дал 1234 паролем(рис. 3.2).

passwd

1234

1234

A terminal window titled 'root@qjarishekkka:~' with search, menu, and close icons in the title bar. The terminal shows a sequence of commands and their outputs: a user switches to root with 'su -', then runs 'passwd'. The prompt changes to '[root@qjarishekkka ~]#'. The user enters 'useradd guest', which results in the message 'useradd: user 'guest' already exists'. Then the user enters 'passwd guest', resulting in 'Changing password for user guest.'. The prompt changes to '[root@qjarishekkka ~]#'. The user enters 'New password:', which results in 'BAD PASSWORD: The password is shorter than 8 characters'. The user enters 'Retype new password:', which results in 'passwd: all authentication tokens updated successfully.'. The prompt returns to '[root@qjarishekkka ~]#'.

```
[qjarishekkka@qjarishekkka ~]$ su -
Password:
[root@qjarishekkka ~]# useradd guest
useradd: user 'guest' already exists
[root@qjarishekkka ~]# passwd guest
Changing password for user guest.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@qjarishekkka ~]#
```

Рис. 3.2: изменение пароля

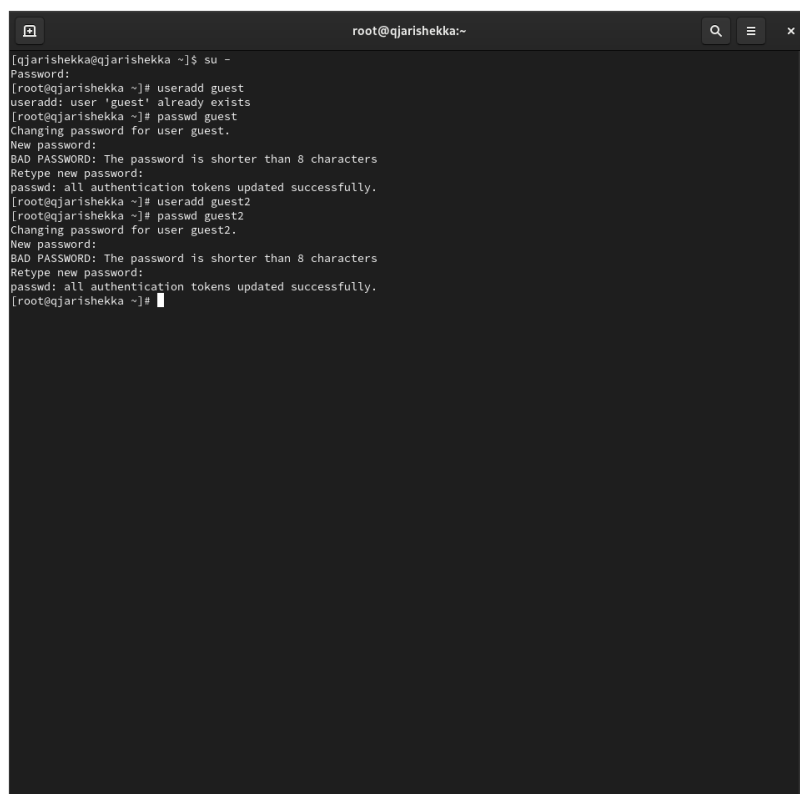
Дальше я повторил шаги 1 2 чтобы создать еще раз нового пользователя (рис. 3.3).

useradd guest2

passwd

1234

1234

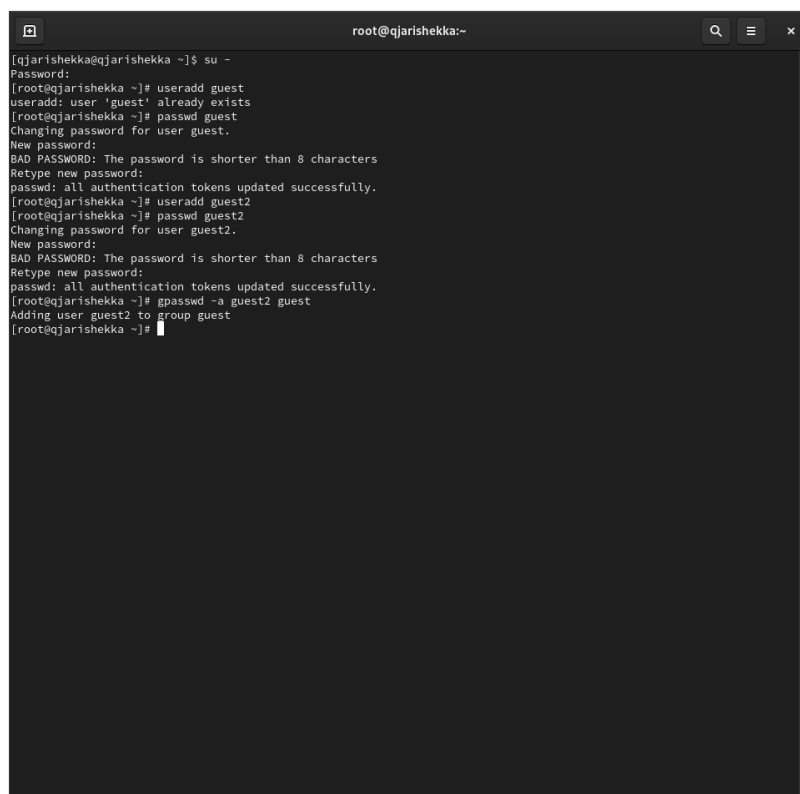
A terminal window titled 'root@qjarishekkka:~' showing the process of creating a new user. The user runs 'su -' to become root. Then they run 'useradd guest', which fails because the user 'guest' already exists. They then run 'passwd guest' to set a password for 'guest', which fails because the password is too short. They then run 'useradd guest2', which succeeds. Finally, they run 'passwd guest2' to set a password for 'guest2', which also fails because the password is too short. The terminal shows the following commands and output:

```
[qjarishekkka@qjarishekkka ~]$ su -
Password:
[root@qjarishekkka ~]# useradd guest
useradd: user 'guest' already exists
[root@qjarishekkka ~]# passwd guest
Changing password for user guest.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@qjarishekkka ~]# useradd guest2
[root@qjarishekkka ~]# passwd guest2
Changing password for user guest2.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@qjarishekkka ~]#
```

Рис. 3.3: Создание пользователя guest2

Потом я добавил пользователя guest2 в группу guest (рис. 3.4).

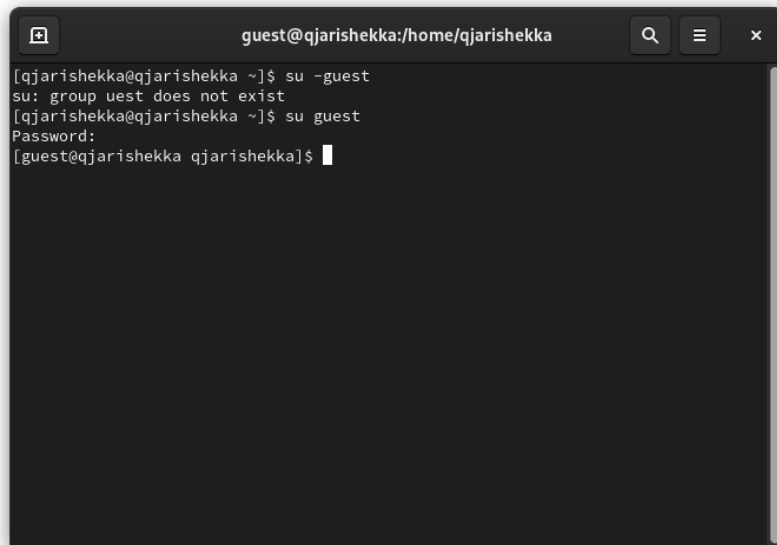
```
gpasswd -a guest2 guest
```

A terminal window titled 'root@qjarishekkka:~' showing a series of commands and their outputs. The user 'root' runs 'su -' to become the root user. Then, 'useradd guest' is run, which succeeds. Next, 'passwd guest' is run to set a password for 'guest'. The password is rejected as being shorter than 8 characters. After retyping, the password is accepted and authentication tokens are updated. Then, 'useradd guest2' is run, which also succeeds. 'passwd guest2' is run, and the password is again rejected as being too short. After retyping, it is accepted and authentication tokens are updated. Finally, 'gpasswd -a guest2 guest' is run to add 'guest2' to the 'guest' group, which succeeds. The terminal ends with the root prompt.

```
root@qjarishekkka:~]$ su -
Password:
[root@qjarishekkka ~]# useradd guest
useradd: user 'guest' already exists
[root@qjarishekkka ~]# passwd guest
Changing password for user guest.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@qjarishekkka ~]# useradd guest2
[root@qjarishekkka ~]# passwd guest2
Changing password for user guest2.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@qjarishekkka ~]# gpasswd -a guest2 guest
Adding user guest2 to group guest
[root@qjarishekkka ~]#
```

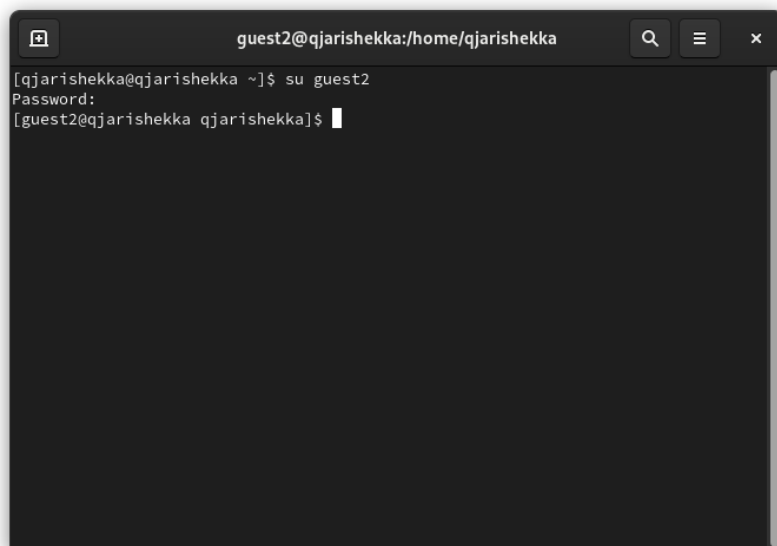
Рис. 3.4: добавление пользователя в группу guest

Затем я вошел в систему от двух пользователей на двух разных консолях рис. 3.5 и рис. 3.6.



```
guest@qjarishekkka:/home/qjarishekkka
[qjarishekkka@qjarishekkka ~]$ su -guest
su: group uest does not exist
[qjarishekkka@qjarishekkka ~]$ su guest
Password:
[guest@qjarishekkka qjarishekkka]$
```

Рис. 3.5: терминал с пользователем guest



```
guest2@qjarishekkka:/home/qjarishekkka
[qjarishekkka@qjarishekkka ~]$ su guest2
Password:
[guest2@qjarishekkka qjarishekkka]$
```

Рис. 3.6: терминал с пользователем guest2

Потом я выполнил команду `pwd` чтобы определить директории в которых находился по каждому терминалу рис. 3.7 и рис. 3.8.

`pwd`

`pwd`



Рис. 3.7: директория первого терминала



Рис. 3.8: директория второго терминала

я отметил что директории разные, это бывает потому что у разных пользователей есть разные каталоги, которые они контролируют.

Потом я уточнил информацию о пользователях от рис. 3.9 до рис. 3.13.

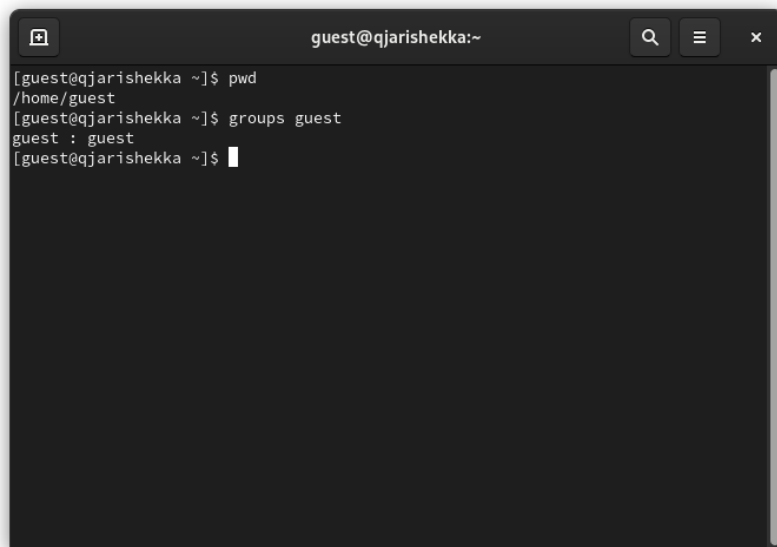
```
groups guest
```

```
groups guest2
```

```
groups
```

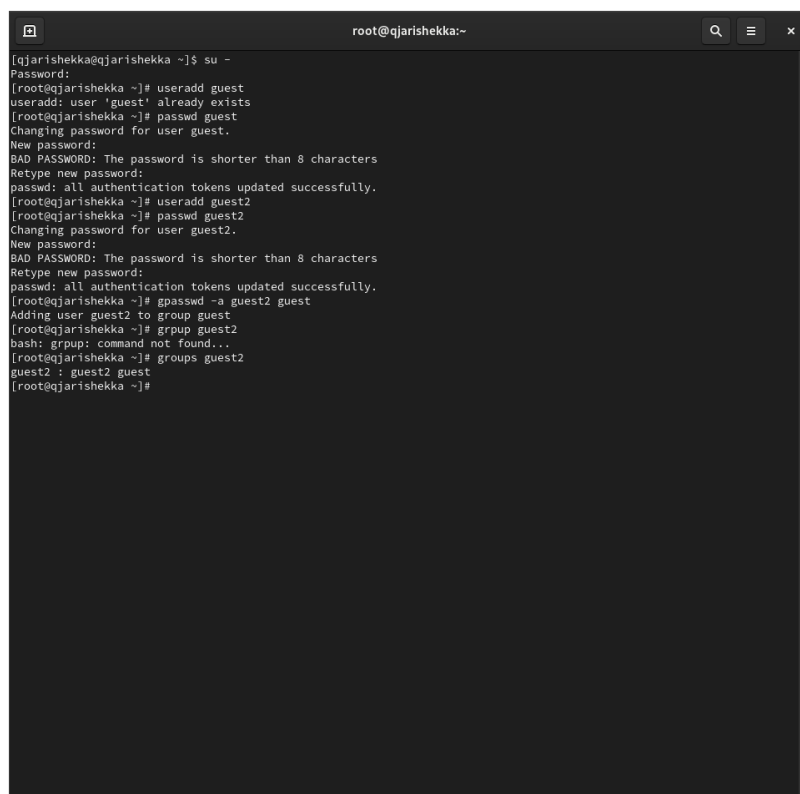
```
id -Gn
```

```
id -G
```

A terminal window titled 'guest@qjarishekk:~' with search, menu, and close buttons. The terminal shows the following commands and output:

```
[guest@qjarishekk ~]$ pwd
/home/guest
[guest@qjarishekk ~]$ groups guest
guest : guest
[guest@qjarishekk ~]$
```

Рис. 3.9: группа



```
root@qjarishekkka:~  
[qjarishekkka@qjarishekkka ~]$ su -  
Password:  
[root@qjarishekkka ~]# useradd guest  
useradd: user 'guest' already exists  
[root@qjarishekkka ~]# passwd guest  
Changing password for user guest.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@qjarishekkka ~]# useradd guest2  
[root@qjarishekkka ~]# passwd guest2  
Changing password for user guest2.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@qjarishekkka ~]# gpasswd -a guest2 guest  
Adding user guest2 to group guest  
[root@qjarishekkka ~]# grpup guest2  
bash: grpup: command not found...  
[root@qjarishekkka ~]# groups guest2  
guest2 : guest2 guest  
[root@qjarishekkka ~]#
```

Рис. 3.10: группа guest2

```
root@qjarishekkka:~  
[qjarishekkka@qjarishekkka ~]$ su -  
Password:  
[root@qjarishekkka ~]# useradd guest  
useradd: user 'guest' already exists  
[root@qjarishekkka ~]# passwd guest  
Changing password for user guest.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@qjarishekkka ~]# useradd guest2  
[root@qjarishekkka ~]# passwd guest2  
Changing password for user guest2.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@qjarishekkka ~]# gpasswd -a guest2 guest  
Adding user guest2 to group guest  
[root@qjarishekkka ~]# grpup guest2  
bash: grpup: command not found...  
[root@qjarishekkka ~]# groups guest2  
guest2 : guest2 guest  
[root@qjarishekkka ~]# groups  
root  
[root@qjarishekkka ~]#
```

Рис. 3.11: группа моего ползователя

```
root@qjarishekkka:~  
[qjarishekkka@qjarishekkka ~]$ su -  
Password:  
[root@qjarishekkka ~]# useradd guest  
useradd: user 'guest' already exists  
[root@qjarishekkka ~]# passwd guest  
Changing password for user guest.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@qjarishekkka ~]# useradd guest2  
[root@qjarishekkka ~]# passwd guest2  
Changing password for user guest2.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@qjarishekkka ~]# gpasswd -a guest2 guest  
Adding user guest2 to group guest  
[root@qjarishekkka ~]# grpup guest2  
bash: grpup: command not found...  
[root@qjarishekkka ~]# groups guest2  
guest2 : guest2 guest  
[root@qjarishekkka ~]# groups  
root  
[root@qjarishekkka ~]# id -Gn  
root  
[root@qjarishekkka ~]#
```

Рис. 3.12: id моего пользователя


```
root@qjarishekkka:~  
[qjarishekkka@qjarishekkka ~]$ su -  
Password:  
[root@qjarishekkka ~]# useradd guest  
useradd: user 'guest' already exists  
[root@qjarishekkka ~]# passwd guest  
Changing password for user guest.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@qjarishekkka ~]# useradd guest2  
[root@qjarishekkka ~]# passwd guest2  
Changing password for user guest2.  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: all authentication tokens updated successfully.  
[root@qjarishekkka ~]# gpasswd -a guest2 guest  
Adding user guest2 to group guest  
[root@qjarishekkka ~]# grpup guest2  
bash: grpup: command not found...  
[root@qjarishekkka ~]# groups guest2  
guest2 : guest2 guest  
[root@qjarishekkka ~]# groups  
root  
[root@qjarishekkka ~]# id -Gn  
root  
[root@qjarishekkka ~]# id -G  
0  
[root@qjarishekkka ~]#
```

Рис. 3.13: номер идентификатора моего пользователя

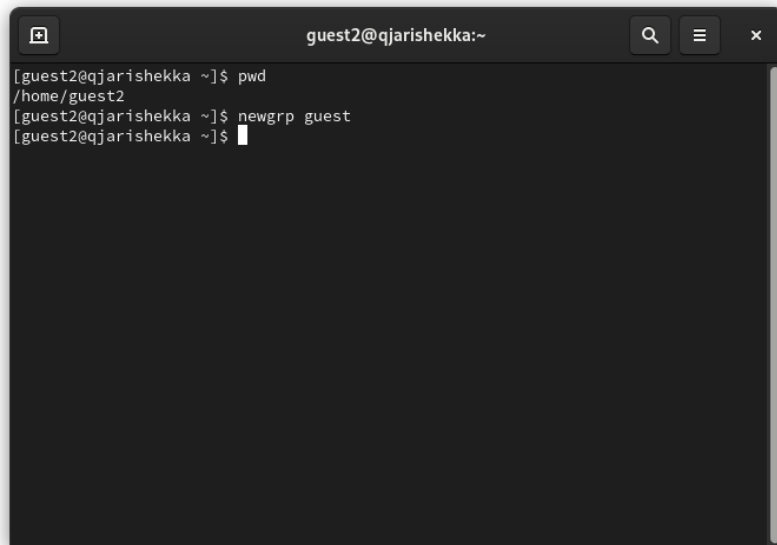
Дальше я сравнил информацию с содержимым файла /etc/group (рис. 3.14).

```
cat /etc/group
```

```
root@qjarishekkka:~  
games:x:20:  
tape:x:33:  
video:x:39:  
ftp:x:50:  
lock:x:54:  
audio:x:63:  
users:x:100:  
nobody:x:65534:  
utmp:x:22:  
utempter:x:35:  
input:x:999:  
kvm:x:36:  
render:x:998:  
systemd-journal:x:190:  
systemd-coredump:x:997:  
dbus:x:81:  
polkitd:x:996:  
avahi:x:70:  
printadmin:x:995:  
ssh_keys:x:101:  
rtkit:x:172:  
pipewire:x:994:  
sssd:x:993:  
sgx:x:992:  
libstoragemgmt:x:991:  
brlapi:x:990:  
tss:x:59:clevis  
geoclue:x:989:  
cockpit-ws:x:988:  
cockpit-wsinstance:x:987:  
flatpak:x:986:  
colord:x:985:  
clevis:x:984:  
setroubleshoot:x:983:  
gdm:x:42:  
stapusr:x:156:  
stapusr:x:157:  
stapdev:x:158:  
pes-ign:x:982:  
gnome-initial-setup:x:981:  
sshd:x:74:  
slocate:x:21:  
chrony:x:980:  
dnsmasq:x:979:  
tcpdump:x:72:  
qjarishekkka:x:1000:  
vboxsf:x:978:  
vboxdrmpc:x:977:  
stapusrpriv:x:159:stapusrpriv  
guest1:x:1001:guest1  
guest2:x:1002:  
[root@qjarishekkka ~]#
```

Рис. 3.14: информация в файле /etc/group

Потом используя терминал guest2 я выполнил команду для регистрации пользователя guest2 в группе guest (рис. 3.15).

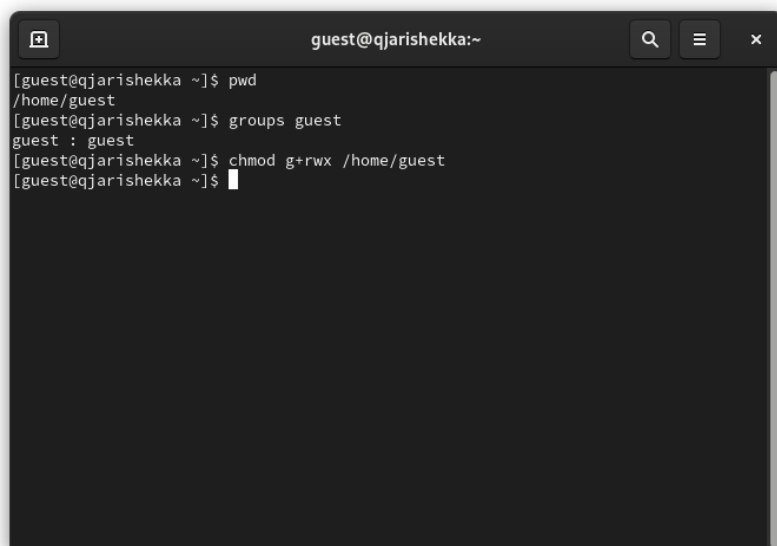


```
guest2@qjarishekk:~  
[guest2@qjarishekk ~]$ pwd  
/home/guest2  
[guest2@qjarishekk ~]$ newgrp guest  
[guest2@qjarishekk ~]$
```

Рис. 3.15: регистрация пользователя guest2 в группе guest

далее я изменил права директории /home/guest используя терминал guest (рис. 3.16).

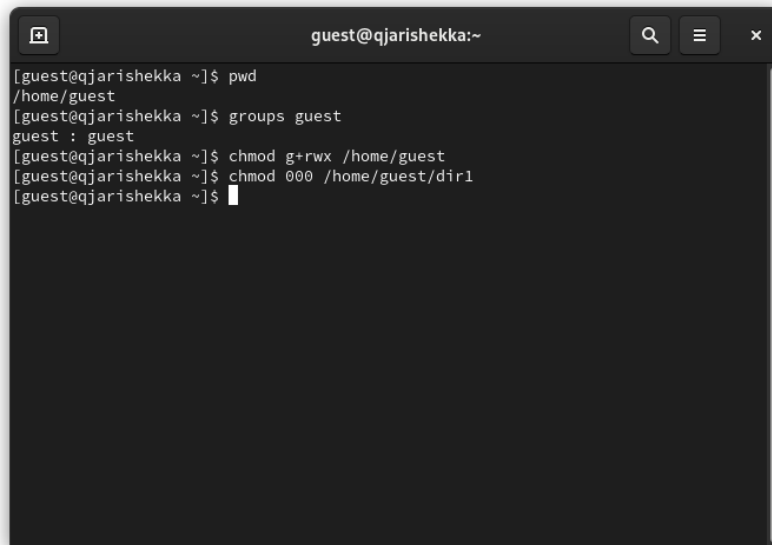
```
chmod g+rx /home/guest
```



```
guest@qjarishekk:~  
[guest@qjarishekk ~]$ pwd  
/home/guest  
[guest@qjarishekk ~]$ groups guest  
guest : guest  
[guest@qjarishekk ~]$ chmod g+rx /home/guest  
[guest@qjarishekk ~]$
```

Рис. 3.16: изменение прав доступа

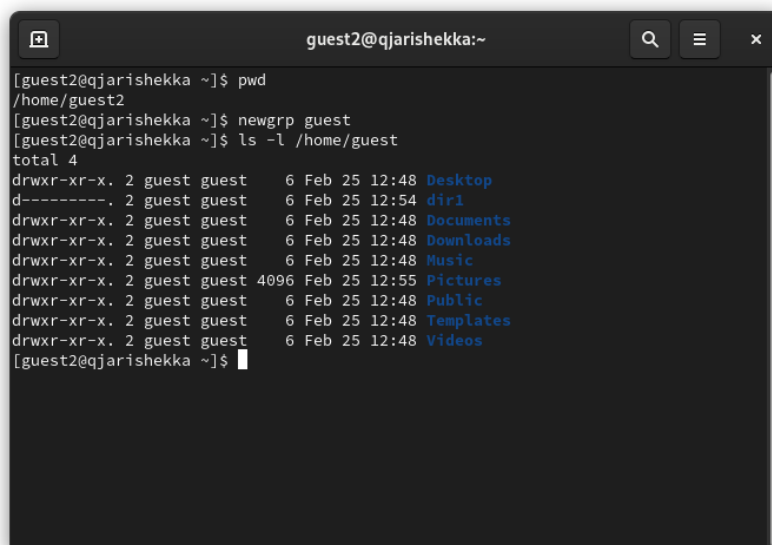
Потом я снял все атрибуты с той же директории (рис. 3.17).



```
guest@qjarishekk:~  
[guest@qjarishekk ~]$ pwd  
/home/guest  
[guest@qjarishekk ~]$ groups guest  
guest : guest  
[guest@qjarishekk ~]$ chmod g+rx /home/guest  
[guest@qjarishekk ~]$ chmod 000 /home/guest/dir1  
[guest@qjarishekk ~]$
```

Рис. 3.17: снятие прав доступа

Потом я проверил правильность снятия атрибутов (рис. 3.18).



```
guest2@qjarishekk:~  
[guest2@qjarishekk ~]$ pwd  
/home/guest2  
[guest2@qjarishekk ~]$ newgrp guest  
[guest2@qjarishekk ~]$ ls -l /home/guest  
total 4  
drwxr-xr-x. 2 guest guest 6 Feb 25 12:48 Desktop  
d------. 2 guest guest 6 Feb 25 12:54 dir1  
drwxr-xr-x. 2 guest guest 6 Feb 25 12:48 Documents  
drwxr-xr-x. 2 guest guest 6 Feb 25 12:48 Downloads  
drwxr-xr-x. 2 guest guest 6 Feb 25 12:48 Music  
drwxr-xr-x. 2 guest guest 4096 Feb 25 12:55 Pictures  
drwxr-xr-x. 2 guest guest 6 Feb 25 12:48 Public  
drwxr-xr-x. 2 guest guest 6 Feb 25 12:48 Templates  
drwxr-xr-x. 2 guest guest 6 Feb 25 12:48 Videos  
[guest2@qjarishekk ~]$
```

Рис. 3.18: проверка

4 Выводы

В эту лабораторную работу я смог вспомнить команды для создания нового пользователя, для регистрации новой группы и также смог уточнить какие действия пользователи могут выполнить над каталогом без прав доступ

Список литературы